

BÁLINT GYEVNÁR

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RESEARCH INTERESTS

Metascience of AI; AI Scientists; Agentic Scientific Discovery; Explainable Multi-Agent Systems; Responsible AI; AI Safety; Human-AI Interaction.

EDUCATION

PhD in Natural Language Processing Sep. 2021 – May 2026

University of Edinburgh

Edinburgh, UK

Thesis: Action Explanation of Multi-Agent Systems via Counterfactual Reasoning

Supervisors: Stefano V. Albrecht, Shay B. Cohen, and Christopher G. Lucas

Integrated Master of Informatics Sep. 2016 – May 2021

University of Edinburgh

Edinburgh, UK

Thesis: Comparison of Account Survival on Twitter During the First Wave of COVID-19

Supervisor: Maria Wolters

Exchange Year in Computer Science Aug. 2018 – Jun. 2019

Nanyang Technological University

Singapore

EXPERIENCE

Postdoctoral Research Associate Sep. 2025 – present

Carnegie Mellon University

Pittsburgh, PA

- Investigating **failure modes of agentic AI scientists**, such as data poisoning, reward hacking, and spin doctoring, with actionable mitigations—work with *Nihar Shah* and *Atoosa Kasirzadeh*.
- Understanding the epistemic **landscape of AI hypothesis generation** and steering, e.g. using formalization of mathematics in Lean—work with *Simon DeDeo* and *David Barack*.

Research Assistant Jul. 2023 – Dec. 2024

University of Edinburgh

Edinburgh, UK

- Researching the intersection of AI safety and AI ethics through content analysis methods—assistant to *Atoosa Kasirzadeh*.
- Large-scale quantitative literature analysis with unsupervised natural language processing tools; curation, topic coding, and qualitative analysis of large corpora of papers.

Teaching Assistant Sep. 2020 – Sep. 2025

University of Edinburgh

Edinburgh, UK

- Tutoring, teaching, marking, and co-supervision in the School of Informatics (see *Teaching*).

Research Intern May 2020 – Oct. 2020

FiveAI

Edinburgh, UK

- Development and evaluation of goal-based interpretable prediction and planning for autonomous vehicles.
- Main contributor of open-source implementation on GitHub with added support for CARLA.

PUBLIC SERVICE

Academic Reviewing

- *Program committee member*: AAAI 2026, AIES 2025, FAccT 2026, XAI 2025
- *Reviewer*: Nature Communications 2025, Artificial Intelligence Review 2025, International Journal of Human-Computer Interaction 2026, IEEE Transactions on Intelligent Transportation Systems 2025, Journal of Autonomous Agents and Multi-Agent Systems 2026, Academic Mindtrek 2025, CHI 2026, ECAI 2023, EMNLP BlackboxNLP 2025, ICRA 2025, IASEAI 2026, IROS 2025, NeurIPS 2025, NeurIPS AI4Metascience 2026
- *Grant reviewer*: Survival and Flourishing Fund 2025–, UK Research and Innovation 2026

Track Chair, Moral Competence in AI

2026

International Association for Safe & Ethical AI

IASEAI 2026, Paris

Workshop Organiser

Oct. 2025

Evaluating Explainable AI and Complex Decision-Making

ECAI 2025

Vice President

Sep. 2022 – Jun. 2025

Edinburgh University Volleyball Club

Edinburgh, UK

TEACHING

- *Co-supervisor*: two master's students
- *Teaching Assistant (Spring 2024)*: Evaluating Sustainable Lands and Cities; Data, Mobility, Infrastructure
- *Marker (Spring 2022–24)*: Machine Learning Theory; Reinforcement Learning; Natural Language Processing
- *Tutor (Autumn 2020)*: Introductory Applied Machine Learning; Introduction to Computing Systems

AWARDS

- Colours Award for Outstanding Volunteering Contribution to Sports;
Edinburgh University Sports Union; Jun. 2024.
- Love, Sex, and AI — AI100 Early Career Essay Competition Featured Essay;
One Hundred Year Study on Artificial Intelligence at Stanford University; Aug. 2023.
- £4,000 Trustworthy Autonomous Systems Early Career Researcher Award;
UK Research & Innovation; Jul. 2022.
- \$1,000 Shape the Future of ITS Competition Award;
IEEE Intelligent Transportation Systems Society; Jun. 2021.

PUBLICATIONS

* equal contribution

greCAPTCHA: Assessing Understanding as Evidence of Research Authorship Under Generative AI

J. Payan*, B. Gyevnár*, A. Kasirzadeh, N.B. Shah
arXiv preprint (under review at CHI), 2026

Distributed Denial of Science: How Indirect Data Poisoning of AI Systems Can Industrialize Scientific Fraud

B. Gyevnár, A. Kasirzadeh, N.B. Shah
arXiv preprint (under review at Nature), 2026

Action Explanation of Multi-Agent Systems via Counterfactual Reasoning

B. Gyevnár

Doctoral Dissertation, University of Edinburgh, 2026

AI Epistemic Risks: Emerging Mechanisms & Evidence

M. Yang, S. Casper, J. Stray, J. Li, C. Jones, A. Gausen, N. Jaques, B. Christian, B. Gyevnár, H. Kirk, Z. He, D. Zhao, S.S. Looi, J. Levy, K. Hackenburg, E. Seger, M. Kowal, M. Malonza, L. Hewitt, H. Lin, M. Sap, D. Hadfield-Menell, T. Costello, R. Rabbany, J.-F. Godbout, D. Rand, A. Kasirzadeh, G. Pennycook, Y. Bengio, K. Peirine

SSRN preprint, 2026

Integrating Counterfactual Simulations with Language Models for Explaining Multi-Agent Behaviour

B. Gyevnár, C.G. Lucas, S.V. Albrecht, S.B. Cohen

AAMAS 2026

Bridging the Gap in the Responsible AI Divides

B. Gyevnár, A. Kasirzadeh

arXiv preprint (under review at Patterns), 2026

Objective Metrics for Human-Subjects Evaluation in Explainable Reinforcement Learning

B. Gyevnár*, M. Towers*

RLDM 2025

AI Safety for Everyone

B. Gyevnár*, A. Kasirzadeh*

Nature Machine Intelligence, 2025

People Attribute Purpose to Autonomous Vehicles When Explaining Their Behavior: Insights From Cognitive Science for Explainable AI

B. Gyevnár, S. Droop, T. Quillien, S.B. Cohen, N.R. Bramley, C.G. Lucas, S.V. Albrecht

CHI 2025

Explainable AI for Safe and Trustworthy Autonomous Driving: A Systematic Review

A. Kuznietsov*, B. Gyevnár*, C. Wang, S. Peters, S.V. Albrecht

IEEE T-ITS, 2024

Causal Explanations for Sequential Decision-Making in Multi-Agent Systems

B. Gyevnár, C. Wang, C.G. Lucas, S.B. Cohen, S.V. Albrecht

AAMAS 2024

Bridging the Transparency Gap: What Can Explainable AI Learn From the AI Act?

B. Gyevnár, N. Ferguson, B. Schafer

ECAI 2023

Communicative Efficiency or Iconic Learning: Do Communicative and Acquisition Pressures Interact to Shape Colour-Naming Systems?

B. Gyevnár*, G. Dagan*, C. Haley*, S. Guo*, F. Mollica

Entropy, 2022

Deep Reinforcement Learning for Multi-Agent Interaction

I.H. Ahmed, C. Brewitt, I. Carlucho, F. Christianos, M. Dunion, E. Fosong, S. Garcin, S. Guo, B. Gyevnár, T. McInroe, G. Papoudakis, A. Rahman, L. Schäfer, M. Tamborski, G. Vecchio, C. Wang, S.V. Albrecht

AI Communications, 2022

A Human-Centric Method for Generating Causal Explanations in Natural Language for Autonomous Vehicle Motion Planning [best paper runner-up]

B. Gyevnár, M. Tamborski, C. Wang, C.G. Lucas, S.B. Cohen, S.V. Albrecht

IJCAI 2022 Workshop on Artificial Intelligence for Autonomous Driving

GRIT: Fast, Interpretable, and Verifiable Goal Recognition with Learned Decision Trees for Autonomous Driving

C. Brewitt, B. Gyevnár, S. Garcin, S.V. Albrecht
IROS 2021

Interpretable Goal-based Prediction and Planning for Autonomous Driving

S.V. Albrecht, C. Brewitt, J. Wilhelm, B. Gyevnár, F. Eiras, M. Dobre, S. Ramamoorthy
ICRA 2021

INVITED TALKS

AI in Scientific Research: What Could Go Wrong and What To Do About It?

Machine Learning Summer School, Columbia University, 2026.

Human and AI Solution Paths in Formalizing Expert Mathematics

Workshop on Social Reasoning and the Ecology of Thought, IVADO & Mila, Montréal, 2026.

AI Safety for Everyone

Ninth Workshop of the Center for Human-Compatible AI, UC Berkeley, 2025.

How Do We Make Explainable AI Work for People?

Machine Learning and Modelling Seminar, Charles University, Prague; and Czech Technical University, Prague, 2024.

MEDIA COVERAGE

- *greCAPTCHA* (2026): TMLR Blog; Times Higher Education
- *Distributed Denial of Science* (2026): Nature News, “Doctored data sets could trick AI agents”